Daniel Duhnev

Machine Learning Engineer

github.com/daniel-duhnev
daniduhnev@gmail.com
linkedin.com/in/daniel-duhnev
daniel-duhnev.github.io

ABOUT

- ML Engineer with an MSc in AI and 5+ years engineering experience in industry.
- Hands-on with PyTorch, Computer Vision, and Deep Learning methods across 2 research placements.
- Built GenAI and Agentic systems with RAG, LLM routing, and MCP integrations at Alexion and Mango.

EDUCATION

MSc Erasmus Mundus in Artificial Intelligence

Universitat Pompeu Fabra / Radboud University

📅 2024 - 2026

- One of 25 Erasmus Mundus scholarship recipients with >2,000 applicants.
- Research placement at Radboudumc: Explored how spatial and intensity data augmentations affect nnU-Net v2 performance on Universal Lesion Segmentation. (GitHub)
- Thesis: "Radiomics-Based Analysis of Grayscale Ultrasound for Pancreas Transplant Rejection Detection" (e-Repository UPF)

BSc Physics

University College London

📅 2018 - 2021

- Awarded First Class Honours, ranking in the top 10% of the class.
- Applied Mathematics, Statistics & Data Analysis; Computational Physics; Practical Machine Learning for Physics
- Python for Scientific Computing - NumPy, SciPy, pandas, matplotlib

SKILLS

ML: PyTorch | scikit-learn | Computer Vision | Deep Learning

GenAI: Requesty (LLM routing) | RAG | MCP | Agno

Data and Cloud: GCP | AWS | Kubernetes | Python | Bash | Databricks

Languages: English | Spanish | Bulgarian

EXPERIENCE

Senior MLOps Engineer

Mango

📅 06/2026 - Present 📌 Barcelona, Spain

Mango is a global fashion retailer operating in 120 markets, built around its e-commerce platform at shop.mango.com.

- Built and maintaining 3 golden path project templates - Airflow DAG batch pipelines, FastAPI, and Streamlit apps on EKS. Deployed with Terragrunt via Jenkins CI/CD.
- Collaborating with Pricing, Personalisation, and GenAI teams to standardize training and deployment across Databricks and EKS.
- Integrating GenAI applications with Requesty for LLM routing and AWS Bedrock Knowledge Bases for RAG, powering an internal and client facing chatbot (Lisa), and a content generator for marketing campaign teams.

AI Engineer

Alexion (AstraZeneca Rare Disease)

📅 01/2026 - 06/2026 📌 Barcelona, Spain

Alexion is AstraZeneca's Rare Disease Unit - a global leader in developing therapies for rare and ultra-rare diseases.

- Extended GenIO, Alexion's internal agentic AI framework, with a Snowflake integration via the official open-source MCP server, enabling agents to query enterprise data using natural language.
- Implemented semantic retrieval and Text-to-SQL within agent workflows using Snowflake Cortex Search and Cortex Analyst.

Research Assistant - AI for Clinical Imaging

BCN-MedTech

📅 09/2025 - 12/2025 📌 Barcelona, Spain

BCN-MedTech is UPF's interdisciplinary research unit using medical imaging, machine learning, and related methods to drive personalised medicine.

- Extended the LiverColor platform, aimed at assessing liver graft suitability for transplantation.
- Processed a dataset of bariatric laparoscopic images from Vall d'Hebron and Reus hospitals, adapting the system by designing a pipeline comprising image pre-processing, an autoencoder for unsupervised feature extraction, and a classifier to detect hepatic steatosis.

Site Reliability Engineer

Sky

📅 10/2021 - 09/2025 📌 London, UK

Sky is Europe's largest media company with over 23 million subscribers.

- Provisioned and maintained GKE & EKS platform for ML and software engineering teams, including deploying Kubeflow Pipelines for recommendation system model training.
- Developed modular IaC using Terraform and Terragrunt, reducing Kubernetes environment setup time from 2 days to 4 hours.
- Managed cloud native resources declaratively using GCP Config Connector and AWS Controllers for Kubernetes (ACK).
- Scaled and stress-tested platform reliability for Super Bowl LVI, handling over 10 million concurrent users with zero outages.
- Cut Kubernetes costs on GKE by automating pod scheduling to the lowest cost nodes, reducing spend by 32% and saving 156,000 USD annually.